# Dynamical System based Socially-aware Navigation for Mobile Robot

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***Abstract*—This paper presents a novel framework for socially-aware robot navigation in a pedestrian mixed environment. To improve the acceptance of mobile robots in daily life, robots should exhibit safe and socially acceptable motions to humans, as integrating social norms has been shown to contribute to comfort and socially-aware behavior in various studies. In this paper, we propose a novel framework, Socially-aware Dynamical System based Control (Social-DSC), which leverages the Dynamical System based Control (DSC) for collision avoidance and Social Force Model (SFM) for socially-aware robot navigation. While safety is guaranteed by the DSC framework, the social norms are considered in the modified SFM. In simulation studies, we compared the proposed method with baseline methods, demonstrating higher pedestrian path smoothness and reduced effort in evading motion, thereby improving human comfort.**

***Index Terms*—Mobile Robots, Socially-aware Navigation, Dynamical Systems based Control, Social Force Model, Collision Avoidance**

## I. Introduction

With the increasing demand for service and assistive robots, it becomes essential for these robots to navigate in a socially-acceptable manner. While safety is the primary concern during robot navigation around humans, human psychological aspects should be considered to enhance human acceptance of mobile robots, necessitating the development of Socially-aware Navigation (SAN). Recent survey works [1], [2] indicate that integrating human-robot interaction (HRI) into robot motion planning is crucial for SAN.

Existing approaches to SAN can be broadly categorized into learning-based and rule-based methods. Learning-based methods leverage machine learning-based techniques to model HRI. For example, in deep learning based methods [3], [4], reward functions that encode socially-aware terms are used to train the robot control policy. The trained models can capture complex and stochastic pedestrian behavior, but successful training requires large and diverse datasets or simulators that generate human trajectories using rule-based approaches [3], and the lack of interpretability of the resulting model makes it difficult to explicitly study the collision avoidance behavior.

Rule-based approaches model HRI through predefined rules and integrate them into traditional navigation algorithms. Proxemics theory [5] is one of the popular concepts of social norm that studies human spatial behavior in social contexts. In robotics, to respect social distance, the proxemic zone is often modeled using potential functions, and an optimization-based method is applied to find a suitable robot trajectory, such as A* [6] or Timed Elastic Band [7]. Many navigation frameworks based on proxemics focus on scenarios where humans are stationary, and the computational speed of optimization methods may be insufficient for dynamic situations.

The Social Force Model (SFM) [8] is another widely studied method that models interactions between pedestrians and their environment as interaction forces. In [9], SFM has been extended to incorporate interaction between human and robot, and the force parameters are identified using experiment data. [10] utilize the geometrical information of the environment to calculate a preferred gap for robot navigation and incorporate it into the SFM. [11] extended SFM with the robot's physical touch force to interact with the humans to solve the freezing robot problem. The challenge of deploying SFM to robots lies in the necessity of carefully calibrating hand-designed parameters to accommodate different scenarios [9]. Moreover, while SFM implicitly reflects safety distance through repulsive force, it does not inherently guarantee collision avoidance [12]. Recently, SFM-based HRI models have been embedded into various frameworks for socially-aware navigation, such as A* [12], RRT [13] and dynamic window approach [11].

We propose the Social-DSC, a novel framework combining the principles of SFM with Dynamical Systems based Control (DSC). DSC is a closed-form approach for robot collision avoidance, which ensures safety and convergence to goal positions [14], [15]. The capability of DSC for collision avoidance has been qualitative evaluated in pedestrian-rich environments [14], [16]. Notably, our work extends DSC by incorporating SFM to influence the robot motion, and we evaluate the proposed framework by comparing it with the baseline methods in SAN. To our knowledge, the application of DSC in the SAN context has not been previously studied.

This paper is organized as follows. Section III covers the problem setup and the preliminaries of the DSC collision avoidance. Section IV introduces the proposed Social-DSC method. Section V presents our simulation results and Section VI concludes with remarks.

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## II. Notations

$f(a;b)$ denotes a function of variable $a$ and parameter $b$. For a 2D vector $\boldsymbol{w} = [w_x, w_y]^T$, we redefine the atan2 function with the vector as the input: $\text{atan2}(\boldsymbol{w}) = \text{atan2}(w_y, w_x)$. $\langle \boldsymbol{w}, \boldsymbol{v} \rangle$ denotes the dot product of vectors $\boldsymbol{w}$ and $\boldsymbol{v}$. The cross product of two 2D vectors $\boldsymbol{a} = [a_x, a_y]^T, \boldsymbol{b} = [b_x, b_y]^T$ is defined as $\boldsymbol{a} \times \boldsymbol{b} = a_x b_y - b_x a_y$.

## III. Background

### A. Problem Statement

Consider a mobile robot $r$, which follows its nominal dynamics $\boldsymbol{f}(\boldsymbol{\xi}_r) : \mathbb{R}^2 \mapsto \mathbb{R}^2$ to accomplish a navigation task:

$$\dot{\boldsymbol{\xi}}_r = \boldsymbol{f}(\boldsymbol{\xi}_r) \tag{1}$$

where $\boldsymbol{\xi}_r(t) \in \mathbb{R}^2$ denotes the robot position at time $t$. We assume that there is a high-level trajectory planner that calculates the nominal dynamics $\boldsymbol{f}(\boldsymbol{\xi}_r)$ for the robot to guide it towards the navigation goal point in the absence of pedestrians.

When encountering unexpected obstacles and pedestrians, the robot should replan its motion while keeping the motion as close as possible to the nominal dynamics. By means of the DSC based collision avoidance algorithm [14], the robot control input can be computed as

$$\dot{\boldsymbol{\xi}}_r = \boldsymbol{M}(\boldsymbol{\xi}_r)\boldsymbol{f}(\boldsymbol{\xi}_r) \tag{2}$$

where $\boldsymbol{M}(\boldsymbol{\xi}_r) \in \mathbb{R}^{2\times 2}$ denotes the modulation matrix, detailed in Section III-B. However, DSC depends solely on the geometric relationships between nominal dynamics and the obstacle shapes. In addition, social awareness needs to be considered to avoid human discomfort. This work aims to develop an algorithm that adapts the nominal dynamics $\boldsymbol{f}(\boldsymbol{\xi}_r)$ in Eq. 1, ensuring both safety and human comfort.

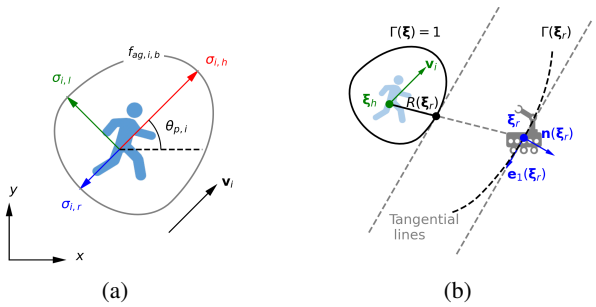
### B. The DSC based Collision Avoidance

The safety zone around a static obstacle are described using common convex shapes, such as circle, ellipse or rectangular. For pedestrians, the proxemics theory is used to model the proxemic zone, see Fig. 1a and Section IV-A. Denote subscript $\star \in \{o, p\}$ to represent an obstacle $o$ or a pedestrian $p$. Within the DSC framework, the obstacle safety zones and pedestrian proxemic zones can be described by the distance function $\Gamma_\star(\boldsymbol{\xi}) : \mathbb{R}^2 \mapsto \mathbb{R}_{>0}$:

$$\Gamma_\star(\boldsymbol{\xi}) = \left( \frac{\|\boldsymbol{\xi} - \boldsymbol{\xi}_\star\|}{R(\tilde{\boldsymbol{\xi}})} \right)^2, \forall \boldsymbol{\xi} \neq \boldsymbol{\xi}_\star \tag{3}$$

where $\boldsymbol{\xi}_\star$ is the position of the obstacle[1] $R(\tilde{\boldsymbol{\xi}})$ is the local radius, defined as the distance from $\boldsymbol{\xi}_\star$ to the obstacle boundary point where $\Gamma_\star(\boldsymbol{\xi}) = 1$, measured in the direction of $\tilde{\boldsymbol{\xi}} = \boldsymbol{\xi} - \boldsymbol{\xi}_\star$(see Fig. 1b). Formally, the obstacle boundary is defined by the set $\mathcal{X}_\star^b = \{\boldsymbol{\xi} \in \mathbb{R}^2 : \Gamma_\star(\boldsymbol{\xi}) = 1\}$, and the boundary divides the space into two regions: the points exterior to the boundary $\mathcal{X}_\star^e = \{\boldsymbol{\xi} \in \mathbb{R}^2 : \Gamma_\star(\boldsymbol{\xi}) > 1\}$ and the points inside the boundary $\mathcal{X}_\star^i = \{\boldsymbol{\xi} \in \mathbb{R}^2 : \Gamma_\star(\boldsymbol{\xi}) < 1\}$. For clarity, we will hereafter omit the subscript $\star$ of the variables when the context is clear.

[1]Due to the unified modeling of proxemic zones and obstacles using Eq. 3, we refer to both as 'obstacle'.




(a) (b)

Fig. 1: (a) Modeling of proxemic zone around a pedestrian using the equipotential contour of the asymmetric Gaussian potential field, parameterized by the variances $\sigma_{i,h}, \sigma_{i,l}, \sigma_{i,r}$. (b) Construction of the basis matrix. The solid black curve represents the boundary of the proxemic zone around the pedestiran $i$, and the dashed black arc shows a part of the level line of $\Gamma(\boldsymbol{\xi}_r)$. The grey lines are the tangents evaluated at robot position.

For the robot at position $\boldsymbol{\xi}_r \in \mathcal{X}^e$, following the nominal dynamics $\boldsymbol{f}(\boldsymbol{\xi}_r)$ and encountering an obstacle described by $\Gamma(\boldsymbol{\xi})$, the normal vector $\boldsymbol{n}(\boldsymbol{\xi}_r)$ to the boundary of $\Gamma(\boldsymbol{\xi}_r)$ can be calculated by normalizing the gradient of $\Gamma(\boldsymbol{\xi}_r)$. A tangent vector $\boldsymbol{e}(\boldsymbol{\xi}_r)$ can be defined to form an orthogonal basis matrix $\boldsymbol{E}(\boldsymbol{\xi}_r)$ with $\boldsymbol{n}(\boldsymbol{\xi}_r)$ at point $\boldsymbol{\xi}_r$ (see Fig. 1b).:

$$\boldsymbol{E}(\boldsymbol{\xi}_r) = \left[\boldsymbol{n}(\boldsymbol{\xi}_r), \boldsymbol{e}(\boldsymbol{\xi}_r)\right] \tag{4}$$

The modulation matrix in Eq. 2 is obtained as

$$\boldsymbol{M}(\boldsymbol{\xi}_r) = \boldsymbol{E}(\boldsymbol{\xi}_r)\boldsymbol{D}(\boldsymbol{\xi}_r)\boldsymbol{E}(\boldsymbol{\xi}_r)^{-1} \tag{5}$$

$$\boldsymbol{D}(\boldsymbol{\xi}_r) = diag(\lambda_n(\boldsymbol{\xi}_r), \lambda_e(\boldsymbol{\xi}_r)) \tag{6}$$

Eq. 5 and 6 explain the concept of DSC: the nominal dynamics $\boldsymbol{f}(\boldsymbol{\xi}_r)$ is first transformed into the basis frame established by $\boldsymbol{E}(\boldsymbol{\xi}_r)$, then modulated by the eigenvalues of the matrix $\boldsymbol{D}(\boldsymbol{\xi}_r)$ along the normal and tangential directions. Finally, the modulated velocity is transformed to the world frame. Collision avoidance is achieved by ensuring that the eigenvalue $\lambda_n(\boldsymbol{\xi}_r)$ vanishes at the boundary point [14]. The eigenvalues can be designed as

$$\lambda_n(\boldsymbol{\xi}_r) = \lambda_v + (1 - \lambda_v)(1 - \frac{1}{\Gamma(\boldsymbol{\xi}_r)}) \tag{7}$$

$$\lambda_e(\boldsymbol{\xi}_r) = \lambda_v + (1 - \lambda_v)(1 + \frac{1}{\Gamma(\boldsymbol{\xi}_r)}) \tag{8}$$

$$\lambda_v(\boldsymbol{\xi}_r, \boldsymbol{v}_r; \boldsymbol{\xi}_\star, \boldsymbol{v}_\star) = \frac{1}{1 + \exp(-10\,\langle \boldsymbol{v}_r - \boldsymbol{v}_\star, \boldsymbol{\xi}_r - \boldsymbol{\xi}_\star \rangle)} \tag{9}$$

where $\lambda_v$ is a sigmoid function to negate the "tail effect", as discussed in [15]. When the robot approaches the obstacle, $\lambda_v \approx 0$, allowing modulation to take effect. When robot moves away from the obstacle, $\lambda_v \approx 1$, causing the modulation matrix to become an identity matrix. This ensures that the robot does not conform to the shape of the obstacle when moving away, thereby maintaining a natural and efficient motion.

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## IV. The Social-DSC Method

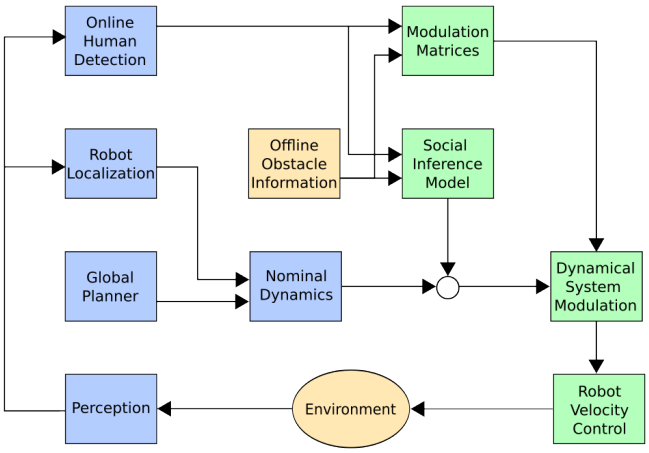



Fig. 2: Overview of the proposed Social-DSC framework. Green blocks represent the content proposed in this work. Orange blocks represent environment information. Blue blocks are outside the scope of this work.

Figure 2 shows the architecture of the proposed Social-DSC method. The input to the Social-DSC is the nominal dynamics $\boldsymbol{f}(\boldsymbol{\xi}_r)$, determined by the Global Planner block. Inspired by the formulation in [15], the Social-DSC is given as:

$$\dot{\boldsymbol{\xi}}_r = \boldsymbol{M}_{safety}(\boldsymbol{\xi}_r)\left(\boldsymbol{f}(\boldsymbol{\xi}_r) + \boldsymbol{f}_{SIM}(\boldsymbol{\xi}_r)\right) \tag{10}$$

The concept of Social-DSC can be explained as the following steps. First, the modulation matrice $\boldsymbol{M}_{safety}(\boldsymbol{\xi}_r)$ is computed for individual obstacles and the proxemic zones of pedestrians (Section IV-B). Then, an additional dynamics $\boldsymbol{f}_{SIM}(\boldsymbol{\xi}_r)$ is generated by the Social Inference Model block which generates a velocity vector that guides the robot's movement in a manner that aligns with social behaviors (Section IV-C). Finally, the Eq. 10 is applied to ensure that the resulting motion avoids collisions with obstacles and maintains a socially acceptable distance from pedestrians.

### A. Modeling of Human Proxemic Zone

To integrate human spatial behavior into DSC framework, we utilize the proxemics theory, which formalizes the space around a pedestrian into four zones: intimate zone, personal zone, social zone and public zone [5]. The social zone refers to the distance that people maintain during social interactions to respect for personal space. In [17], this zone is defined to range from 1.22m to 3.66m for human-robot interaction.

We model the proxemic zone of pedestrian $i$ as an equi-potential contour of the **a**symmetric **G**aussian potential field [18] $f_{ag,i}(\boldsymbol{\xi}_r; \boldsymbol{\xi}_{p,i}, \theta_{p,i}, \sigma_{i,h}, \sigma_{i,l}, \sigma_{i,r}) : \mathbb{R}^2 \mapsto \mathbb{R}_{>0}$ (Fig. 1a). The parameters $\boldsymbol{\xi}_{p,i}, \theta_{p,i}$ denote the position and orientation of the pedestrian, and $\sigma_{i,h}$, $\sigma_{i,l}$, $\sigma_{i,r}$ are the variances along the heading, lateral and rear directions, respectively. The proxemic zone is defined as the region bounded by the level line $f_{ag,i}(\boldsymbol{\xi}_{p,i} + \sigma_{i,h}\boldsymbol{u}(\theta_{p,i}))$. To balance between robot motion efficiency and human comfort, we empirically choose the values of the variances so that the contour shape represents the personal zone defined in [17], ensuring that the minimum distance between the robot and pedestrians ranges between 1.22m and 3.66m (see Section V). The contour shape is intuitively convex. [2]

### B. Multiple Obstacles

Consider there are $N$ obstacles, the overall modulation matrix $\boldsymbol{M}_{safety}(\boldsymbol{\xi}_r)$ is obtained as the weighted sum of the individual modulation matrices $M_i(\boldsymbol{\xi})$:

$$\boldsymbol{M}_{safety}(\boldsymbol{\xi}_r) = \sum_{i=1}^{N} \omega_i(\boldsymbol{\xi}_r)\boldsymbol{M}_i(\boldsymbol{\xi}_r) \tag{11}$$

$$\omega_i(\boldsymbol{\xi}_r) = \frac{\omega_i'(\boldsymbol{\xi}_r)}{\sum_{j=1}^{N}\omega_j'(\boldsymbol{\xi}_r)} \tag{12}$$

$$\omega_i'(\boldsymbol{\xi}_r) = \prod_{j=1, j\neq i}^{N} (\Gamma_i(\boldsymbol{\xi}_r) - 1) \tag{13}$$

The weighting factors can be interpreted as follows. When the robot is on the boundary of obstacle $i$, the parameters $\omega_j'(\boldsymbol{\xi})$ for other obstacles $j \neq i$ become 0. As a result, the influences from other obstacles $j \neq i$ vanish, and only $\boldsymbol{M}_i$ is active to ensure the impenetrability on obstacle $i$.

### C. Social Inference Model

The Social Inference Model (SIM) leverages the principle of Social Force Model (SFM), which characterizes interactions between humans and objects in the environment through attractive and repulsive forces:

$$\boldsymbol{f}_{SIM} = k_{SIM}\frac{\boldsymbol{F}_{SIM}}{\|\boldsymbol{F}_{SIM}\|} \tag{14}$$

$$\boldsymbol{F}_{SIM} = \omega_{pa}\boldsymbol{F}_{pa} + \omega_{pp}\boldsymbol{F}_{pp} + \omega_{sr}\boldsymbol{F}_{sr} \tag{15}$$

where $k_{SIM}$ is a scaling parameter, $\omega_{pa}, \omega_{pp}$ and $\omega_{sr}$ are the weighting parameters. The forces $\boldsymbol{F}_{pa}, \boldsymbol{F}_{pp}, \boldsymbol{F}_{sr}$ are the forces representing the robot's intentions to engage in **p**edestrian **a**voidance, to ensure **p**riority of **p**edestrian and to follow **s**ocial **r**ules, respectively. Note that while traditional SFM-based approaches apply the resulting force directly to the robot, SIM instead outputs a reference direction of velocity that represents the robot's intent to navigate around pedestrians.

*1) Pedestrian Avoidance:* The influence of the pedestrian $i$ on the robot $r$ can be formulated as a repulsive force:

$$\boldsymbol{f}_{pa,i} = A_{p,i}\exp\left(\frac{d_{p,i} - \|\tilde{\boldsymbol{\xi}}_i\|}{B_{p,i}}\right)\frac{\tilde{\boldsymbol{\xi}}_i}{\|\tilde{\boldsymbol{\xi}}_i\|} \tag{16}$$

$$\tilde{\boldsymbol{\xi}}_i = \boldsymbol{\xi}_r - \boldsymbol{\xi}_{p,i} \tag{17}$$

where $A_{p,i}, B_{p,i}, d_{p,i}$ are the parameters of the forces. For multiple pedestrians, the influence of $P$ pedestrians can be formulated as the sum of the individual forces:

$$\boldsymbol{F}_{pa} = \sum_{i=1}^{P} \boldsymbol{f}_{pa,i} \tag{18}$$

[2]We provide an intuitive introduction to the asymmetric Gaussian function, and suggest the readers to refer to [18] for details.

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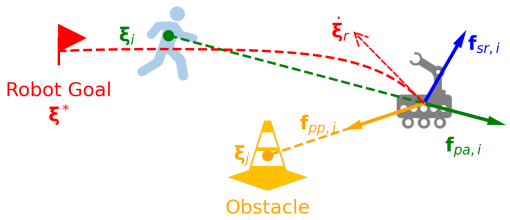



Fig. 3: Social Inference Forces. The red dashed curve represents the nominal dynamics the robot should follow. Dashed lines indicate the interactions between the robot and the pedestrian and the obstacle.

*2) Pedestrian Priority:* It has been found out that human attempt to minimize effort in evading motion [18]. Moreover, robots can cause discomfort if they obstruct human's path. We use the guidance force $\boldsymbol{F}_{pp}$ to steer the robot in a way that reduces interference and effort for the pedestrian.

Assume there are $O$ obstacles obstructing the path between robot $r$ and pedestrian $i$, the inference force $\boldsymbol{f}_{pp,i,j}$ regarding obstacle $j$ and the total force $\boldsymbol{F}_{pp}$ can be formulated as

$$\boldsymbol{f}_{pp,i} = \sum_{j}^{M} \alpha_{r,i}(\boldsymbol{\xi}_j; L) A_{o,j} \exp\left(\frac{d_{o,j} - \|\tilde{\boldsymbol{\xi}}_j\|}{B_{o,j}}\right) \frac{-\tilde{\boldsymbol{\xi}}_j}{\|\tilde{\boldsymbol{\xi}}_j\|} \tag{19}$$

where where $A_{o,j}, B_{o,j}, d_{o,j}$ are the parameters of the forces, $\tilde{\boldsymbol{\xi}}_j = \boldsymbol{\xi}_r - \boldsymbol{\xi}_{o,j}$. The activation function $\alpha_{r,i}(\boldsymbol{\xi}_j; L)$ equals 1 when the obstacle lies within the rectangular formed by the central line segment connecting $\boldsymbol{\xi}_r$ and $\boldsymbol{\xi}_i$ with a width of $L$, and 0 otherwise. For multiple pedestrians, the total force is calculated as:

$$\boldsymbol{F}_{pp} = \sum_{i=1}^{P} \boldsymbol{f}_{pp,i} \tag{20}$$

The definition of $\boldsymbol{F}_{pp}$ diverges from conventional SFM, where forces typically act repulsively to maintain distance between the robot and obstacles. However,it's worth noting that due to the activation function, the force diminishes as the pedestrian moves away from the robot, and the collision avoidance against obstacles is assured by the DSC framework.

*3) Social Rules:* Pedestrians tend to follow conventional social rules to avoid collision. For example, in Europe, people tend to walk on the right-hand side of pathways and corridors. In addition, social conventions need not be considered when the robot is positioned too far to the pedestrian's right-hand side. Authors in [10], [13] have proposed different forms of social forces for social rule. In this work, we use the sigmoid function to model the social rule force regarding pedestrian $i$:

$$\boldsymbol{f}_{sr,i} = \frac{k_{sr,i} \boldsymbol{n}_{v,i,L}}{1 + \exp\left(-B_{sr,i} \lambda_r(\boldsymbol{n}_{v,h}, \tilde{\boldsymbol{\xi}}_i)\right)} \tag{21}$$

$$\lambda_r(\boldsymbol{n}_{v,i}, \tilde{\boldsymbol{\xi}}_i) = \boldsymbol{n}_{v,i} \times \tilde{\boldsymbol{\xi}}_i - \delta_i \tag{22}$$

where $k_{sr,i}, B_{sr,i}$ are the parameters of the sigmoid function, $\boldsymbol{n}_{v,i} = \boldsymbol{v}_i / \|\boldsymbol{v}_i\|$ is the unit vector of pedestrian velocity, $\boldsymbol{n}_{v,i,L}$ is the unit left normal vector to $\boldsymbol{n}_{v,i}$. In Eq. 22, we use the cross product to indicates whether the robot is on the left or right hand side of the pedestrian, and use the offset $\delta_i$ to horizontally shift the sigmoid function, so that the robot considers the social convention when it is not too far away to the pedestrian's right-hand side.

## V. Simulation and Evaluation

We conducted simulations in ROS using pedsim_ros package [19], where the simulated pedestrians are driven by SFM to interact with obstacles, other pedestrians and the robot. We compare Social-DSC with DSC and baseline rule-based methods in SAN: SFM [9], ORCA [20]. An omni-directional mobile robot simulator is used for the evaluation. The simulations are conducted on a PC with an Intel i7-6820HQ CPU and 16GB RAM.

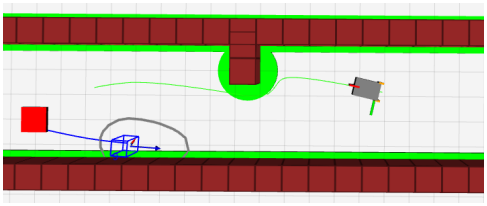


Fig. 4: Visualization of Simulation in Rviz. Pedestrian is shown as blue cubic, robot is shown as grey block. Red square represents the robot goal position, green area represent the safety zone of obstacles and walls. Green curve shows the forward simulation of robot trajectory using Social-DSC.

### A. Simulation Setup

The robot reference velocities are set to $v_{r,ref} = 1.2m/s$, the controllers' sampling time is 0.1s. The SFM controller adopts the identified parameters in [9] for human-robot interaction, and we empirically set $A_{o,i} = 1$ to reduce obstacle influences, while keeping other parameters for obstacle the same as human model. For the ORCA controller, the prediction time horizon is 2s. The radii of robot and pedestrian are set to $0.4m$ and $1m$, respectively, and the robot takes half of the responsibility for collision avoidance. For the DSC, the nominal dynamics is $\boldsymbol{f}(\boldsymbol{\xi}_r) = (\boldsymbol{\xi}^* - \boldsymbol{\xi}_r)$, where $\boldsymbol{\xi}^*$ is a static target position. Based on [18], the parameters are adjusted to $\sigma_h = \Delta T \|\boldsymbol{v}_i\|, \sigma_r = \frac{1}{2}\sigma_h$, where $\|\boldsymbol{v}_p\| = 1.2m/s, \Delta T = 2s$ are the estimated average pedestrian speed and prediction time, respectively. $\sigma_l$ is set to $1.2m$ due to the narrow corridor environment. The Social-DSC uses the same DSC and SFM parameters. $k_{SIM}$ is set to 0.3, $k_{sr,i} = 1$ and $\delta_i = 0.2$. The SIM forces are equally weighted.

The controllers are evaluated in two scenarios: a narrow corridor collision avoidance and a four-agent position-swapping task. In each scenario, agents travel from its start to goal position, with 100 simulation iterations conducted. Pedestrians are driven by default SFM parameters in pedsim_ros, and are randomly spawned within a $1m^2$ area around their initial positions at each iteration. Their velocities are also randomly assigned between $0.26m/s$ and $1.34m/s$ by default in pedsim_ros.

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### B. Metrics

*1) Safety Distance during Interaction ($SD$):* The minimum distance during interaction is measured when the pedestrian is directly in front of the robot and moving towards it.

*2) Path Smoothness ($PS$):* We use Path Smoothness $PS$ to assess the naturalness of motion. A higher $PS$ indicates that the motion contains less sharp turns or large detours. Given the recorded path $\Xi = \{\boldsymbol{\xi}(1), \ldots, \boldsymbol{\xi}(N)\}$ , the $PS$ is calculated as $S = \frac{1}{\sum_{i=1}^{N} \kappa_i}$, where $\kappa_i$ is the curvature at data point $\boldsymbol{\xi}(i)$.

*3) Path Length Ratio ($PR$):* The path length ratio ($PR$) quantifies the effort required by pedestrians and robots to avoid each other: $PR = \frac{D_{actual}}{D_{ref}}$, where $D_{ref}$ is the straight-line distance between the start and end positions, and $D_{actual}$ is the actual path length recorded in simulation.

### C. Simulation Results and Analysis

*1) Narrow Corridor Scenario:* We simulate a $20m \times 4m$ area of our office floor to investigate human-robot interaction in a reduced-scale, enabling focused study of the local motion planners. The pedestrian and the robot travel in opposite directions, starting from opposite ends and aiming to reach the other side. A pillar blocks the path, making it uncomfortable for more than one person to traverse at the same time.

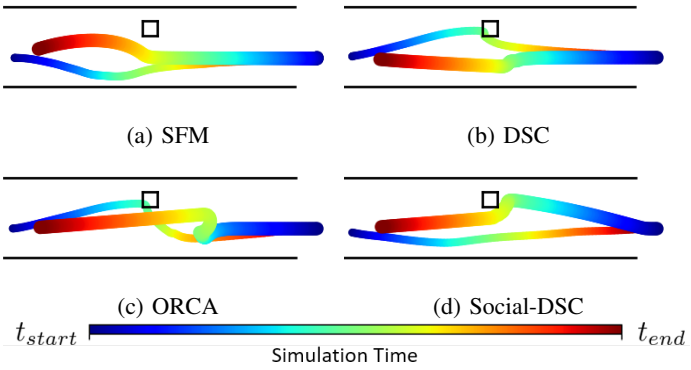



Fig. 5: A set of recorded paths in the corridor simulation. The larger and smaller markers represent the trajectory of the robot and the pedestrian, respectively. The black lines represent the walls and and rectangle shows the pillar on the path.

Fig. 5 shows representative trajectories produced by different controllers. One can see that the pedestrian motion is smoother when the robot is controlled by SFM and Social-DSC. For ORCA, the robot's trajectory is unnatural because the pedestrian does not align with the assumptions made by ORCA. The metrics are visualized as box plots in Fig. 6. In Fig. 6a, SFM and DSC have smaller $SD$ values, while ORCA and Soical-DSC maintain an appropriate social distance of 1.4m. For SFM, the reaction forces are scaled using exponential functions, which primarily manifest when the robot is in close proximity to pedestrians. Similarly, DSM comes into effect near the boundary of the proxemic zone due to the quadratic scaling of the eigenvalues. The higher $SD$s

h

TABLE I: Mean Values of the Metrics in Four Agent Scenario

| Statistic | SFM | DSC | ORCA | Social-DSC |
|---|---|---|---|---|
| Pedestrian 1 | 1.39 | 1.30 | 1.75 | 1.50 |
| Pedestrian 2 | 2.28 | 1.11 | 1.67 | 1.26 |
| Pedestrian 3 | 2.34 | 1.21 | 1.79 | 1.62 |

(a) Minimum Distance to Robot $SD$

| Statistic | SFM | DSC | ORCA | Social-DSC |
|---|---|---|---|---|
| Pedestrian 1 | 2.81 | 2.40 | 2.43 | 2.94 |
| Pedestrian 2 | 7.72 | 5.07 | 3.28 | 5.11 |
| Pedestrian 3 | 8.05 | 5.01 | 2.96 | 6.50 |
| Robot | 2.11 | 1.48 | 0.59 | 6.00 |

(b) Path Smoothness $PS$

| Statistic | SFM | DSC | ORCA | Social-DSC |
|---|---|---|---|---|
| Pedestrian 1 | 1.14 | 1.13 | 1.16 | 1.12 |
| Pedestrian 2 | 1.11 | 1.11 | 1.11 | 1.10 |
| Pedestrian 3 | 1.12 | 1.11 | 1.13 | 1.11 |
| Robot | 1.14 | 1.11 | 1.35 | 1.11 |

(c) Path Ratio $PR$

of ORCA and Social-DSC result from the predictive nature of ORCA and the additional SIM dynamics of Social-DSC.

Fig. 6b compares the $PS$ of the simulated path for pedestrians and robot. Compared to the pedestrians, the robot yields generally lower $PS$, primarily because of the need for larger turning maneuvers around the pillar. Both DSC and Social-DSC yield larger $PS$, because the evading motion depends on the geometric description of obstacles. SFM has lower $PS$ value due to the repulsive nature of the force model.

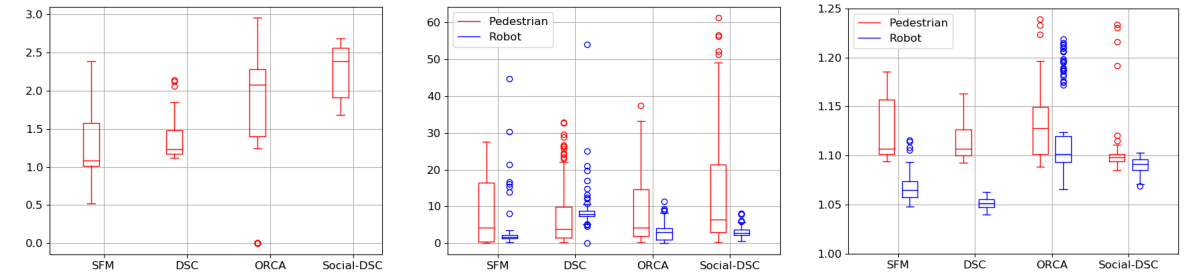
Fig. 6c shows that Social-DSC yield lower pedestrian $PR$ while a higher robot $PR$, which means the robot takes more responsibility in the evading motion. The Path Ratios of ORCA and Social-DSC show that the robot contributes more compared to SFM and DSC, owning to the modeling of the reciprocal behavior. Again, ORCA yields higher pedestrian $PR$, potentially due to pedestrians not adhering to the same reciprocal assumption, leading to trajectory oscillations.

*2) Four Agent Position Swapping:* In this scenario, three pedestrians and a robot are evenly distributed on a circle with a radius of 10m, as shown in Fig. 7. Their goal is to traverse to the opposite side of the circle. The mean values of the metrics are summarized in Table I.

Table IIa shows that all the controllers are capable of maintaining an acceptable social distance during interaction. The Social-DSC yields a larger $SD$ compared to DSC, attributed to the additional evasive maneuvers provided by the SIM module, which considers social awareness.

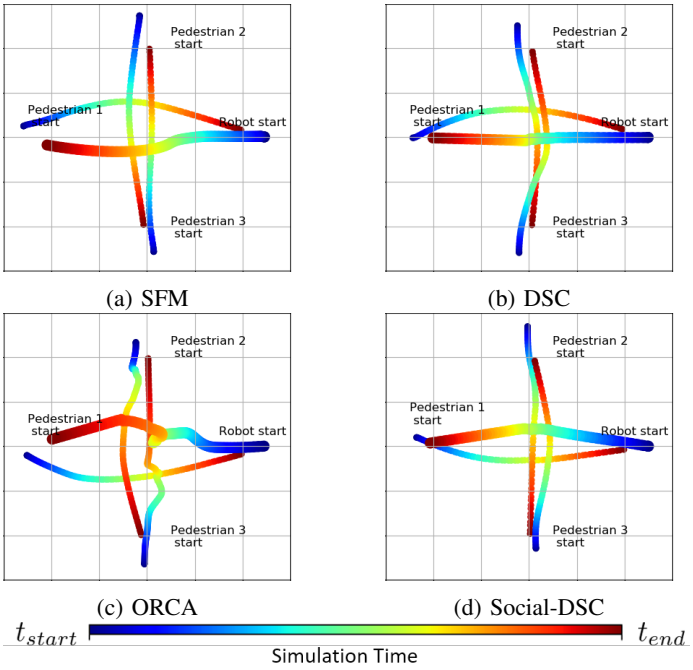
The $PR$ in Table IIc indicate that in this scenario, both pedestrians and the robot exert comparable effort to avoid each other, while robot controlled by ORCA has larger detours, as illustrated in Fig. 7c. This observation is further supported by Table IIb, where the Social-DSC demonstrates better robot $PS$, leading to similar conclusions as drawn in the corridor scenario.

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(a) Minimum Distance During Interaction $SD$ (b) Path Smoothness $PS$ (c) Path Ratio $PR$

Fig. 6: Evaluations of metrics in the Corridor Scenario




(a) SFM (b) DSC (c) ORCA (d) Social-DSC

Fig. 7: A set of recorded paths in the Four Agent Swapping scenario. The larger and smaller markers represent the trajectory of the robot and the pedestrian, respectively.

## VI. Conclusion and Outlooks

In this paper, we present the Soical-DSC, a novel approach that combines the DSC with the SFM to generate safe and socially-aware robot trajectories. Compared to the baseline methods, our approach generates robot motions that explicitly consider the social distance. The proposed method enhances human comfort, as evidenced by the higher smoothness of human paths and a lower path ratio. Future work involve conducting real-world experiments and incorporating adaptive behaviors to enhance the interaction model.

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